Elementary Problems

Arithmetic Function and Floor Function

Version 1.0

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Divisor Function

1. Find the number of positive divisors of each of 20 and 36.
2. Find the sum of all positive divisors of each of 30 and 72.
3. Find all positive integers n for which $d(n) = n$.
4. Find all positive integers n for which $d(n) = n - 1$.
5. Find all positive integers n for which $d(n)$ is odd.
6. Prove that for each positive integer $m \geqslant 2$, there exist infinitely many positive integers n for which $d(n) = m$.
7. Let $\omega(n)$ be the number of distinct prime divisors of n. Prove that there are infinitely many distinct positive integers m and n such that $\omega(m + n) = \omega(m) + \omega(n)$.
8. Let n be an even positive integer. Prove that $\sigma(n) = 2n$ if and only if $n = 2^{p-1}(2^p - 1)$ for some prime $2^p - 1$.

Euler's Totient Function

9. Find the number of positive integers not exceeding 15 which are relatively prime to 15. Find the same thing with 15 replaced by 40.
10. Find the number of positive integers not exceeding 280 which are relatively prime to 28.
11. Prove that $\varphi(n)$ is even for any integer $n \geqslant 3$.
12. Find all positive integers n for which $\varphi(n)$ is not divisible by 4.
13. Find all positive integers n for which $\varphi(n) = 2$.
14. Find all positive integers n for which $\varphi(n) = 4$.
15. Prove that for each positive integer m, there exist finitely many positive integers n for which $\varphi(n) = m$.
16. Prove that $\displaystyle\sum_{d|n} \varphi(d) = n$ for any positive integer n, where the sum runs through all positive divisors d of n.
17. Prove that $\varphi(mn) = \varphi(m)\varphi(n)\dfrac{d}{\varphi(d)}$ for any positive integers m and n where $d = (m, n)$.

18. Find all positive integers m and n such that $\varphi(mn) = \varphi(m) + \varphi(n)$.

Floor Function

19. Find all real numbers x such that
$$\left\lfloor \frac{10}{x} \right\rfloor + \left\lfloor \frac{x}{10} \right\rfloor = 1.$$

20. For any positive integer n, prove that
$$\left\lfloor \frac{n}{\lfloor \sqrt{n} \rfloor + 1} \right\rfloor = \lfloor \sqrt{n} \rfloor \quad \text{or} \quad \lfloor \sqrt{n} \rfloor - 1.$$

21. (Korea MO 1996 Problem 3) Let $a = \lfloor \sqrt{n} \rfloor$ for a given positive integer n. Express the summation $\displaystyle\sum_{k=1}^{n} \lfloor \sqrt{k} \rfloor$ in terms of n and a.

22. Find the value of
$$\sum_{k=1}^{99} \left\lfloor \frac{33k}{100} \right\rfloor.$$

23. (IMO 1968 Problem 6) For every natural number n, evaluate the sum
$$\sum_{k=0}^{\infty} \left\lfloor \frac{n+2^k}{2^{k+1}} \right\rfloor = \left\lfloor \frac{n+1}{2} \right\rfloor + \left\lfloor \frac{n+2}{4} \right\rfloor + \cdots + \left\lfloor \frac{n+2^k}{2^{k+1}} \right\rfloor + \cdots .$$

24. (Romania District Olympiad 2008 Grade 9 Problem 3) Prove that if $n \geqslant 4$, $n \in \mathbb{Z}$ and $\left\lfloor \dfrac{2^n}{n} \right\rfloor$ is a power of 2, then n is also a power of 2.

25. Let $P(x)$ and $Q(x)$ be two non-constant polynomials with real coefficients such that $\lfloor P(x) \rfloor = \lfloor Q(x) \rfloor$ for all real numbers x. Prove that $P = Q$.

Record

v1.0: problems 1–25